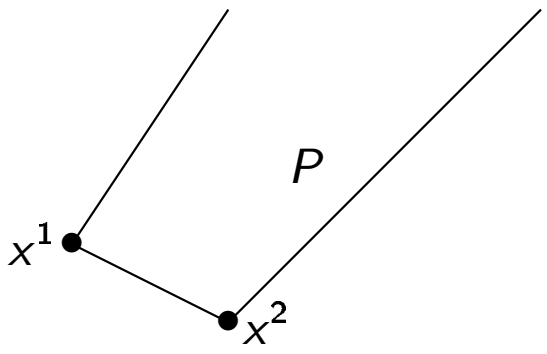
3.3 Basic feasible solutions and vertices of polyhedra

Due to the LP fundamental theorem, to solve any LP it suffices to consider the vertices (finitely many) of the polyhedron P of the feasible solutions.

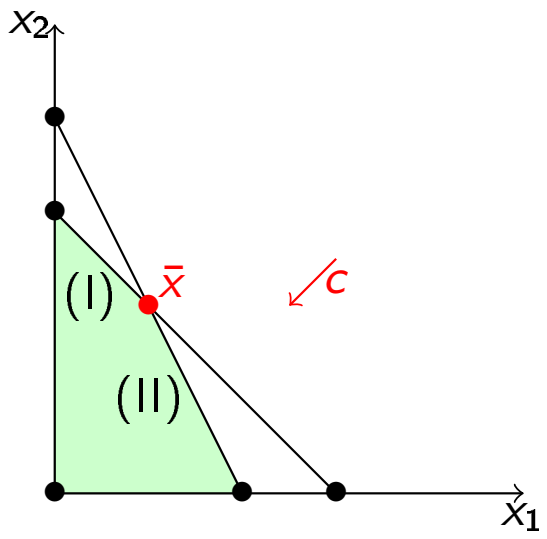


Since the geometrical definition of vertex cannot be exploited algorithmically, we need an algebraic characterization.

Which are the vertices of $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} \leq \underline{b}, \underline{x} \geq \underline{0}\}$ with only inequalities?

Example

$$\begin{array}{lrcll} \min & -x_1 - 3x_2 & & & \\ \text{s. t.} & x_1 + x_2 & \leq & 6 & \text{(I)} \\ & 2x_1 + x_2 & \leq & 8 & \text{(II)} \\ & x_1, x_2 & \geq & 0 & \end{array}$$



A vertex corresponds to the intersection of the hyperplanes associated to n inequalities.

Vertex $\underline{\bar{x}}$ is the intersection of the hyperplanes of (I) and (II), i.e., the solution of the system:

$$\begin{cases} x_1 + x_2 = 6 \\ 2x_1 + x_2 = 8 \end{cases}$$

What about the vertices of polyhedra expressed in standard form?

$$P = \{\underline{x} \in \mathbb{R}^n \ : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$$

We want to solve LPs in standard form. However, these are easier to describe if we start from

$$P = \{\underline{x} \in \mathbb{R}^n : \ A\underline{x} \leq \underline{b}, \underline{x} \geq \underline{0}\},$$

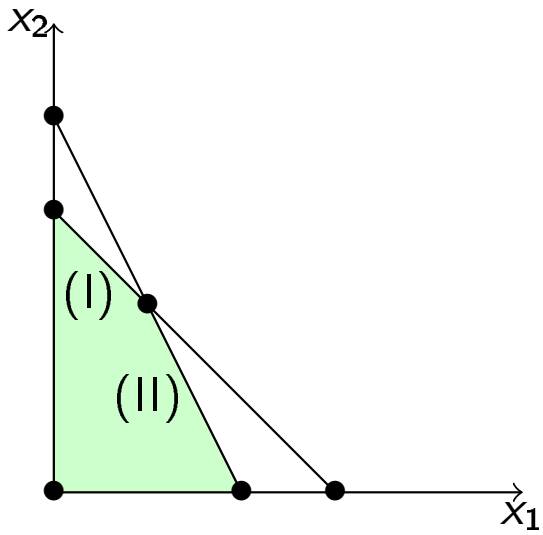
transform it into standard form

$$P' = \{\underline{x} \in \mathbb{R}^n : \ A\underline{x} + \underline{s} = \underline{b}, \underline{x}, \underline{s} \geq \underline{0}\}$$

and rename: $A := [A|I]$ and $\underline{x} := (\underline{x}^T|\underline{s}^T)$, where A has m rows.

Example

$$(P)\quad \begin{array}{rcl} x_1 + x_2 & \leq & 6 \\ 2x_1 + x_2 & \leq & 8 \\ x_1, x_2 & \geq & 0 \end{array} \Rightarrow (P')\quad \begin{array}{rcl} x_1 + x_2 + s_1 & = & 6 \\ 2x_1 + x_2 + s_2 & = & 8 \\ x_1, x_2, s_1, s_2 & \geq & 0 \end{array}$$



Taking the intersection of the lines associated to (I) and (II) in P, amounts in P' to let $s_1 = s_2 = 0$.

Notes:

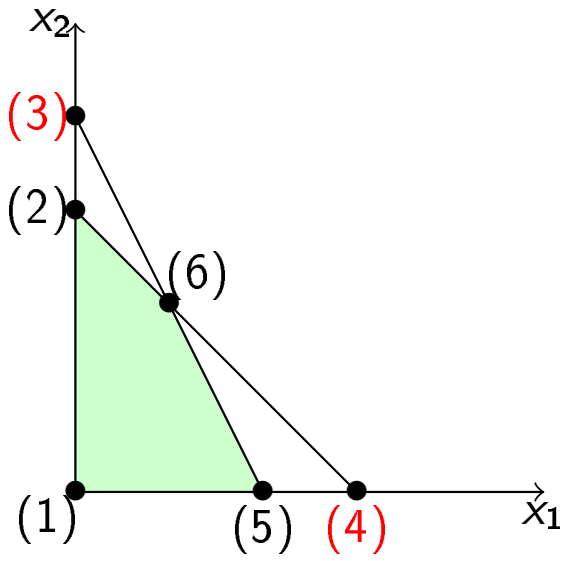
- Every constraint in P corresponds to a slack variable in P'. When $s_i = 0$ the constraint is satisfied with "=".

  E.g. $s_1 = 0 \rightarrow x_1 + x_2 = 6$.
- Any vertex of P is the intersection of the hyperplanes associated to n inequalities. Equivalent to setting the corresponding variables in P' to 0.

Example (continued)

Compute all the intersections

$$\begin{aligned} x_1 + x_2 + s_1 &= 6 \\ 2x_1 + x_2 + s_2 &= 8 \\ x_1, x_2, s_1, s_2 &\geq 0 \end{aligned}$$



(1) $x_1 = 0, x_2 = 0 \Rightarrow s_1 = 6, s_2 = 8$

(2) $x_1 = 0, s_1 = 0 \Rightarrow x_2 = 6, s_2 = 2$

(3) $x_1 = 0, s_2 = 0 \Rightarrow x_2 = 8, s_1 = -2$

(4) $x_2 = 0, s_1 = 0 \Rightarrow x_1 = 6, s_2 = -4$

(5) $x_2 = 0, s_2 = 0 \Rightarrow x_1 = 4, s_1 = 2$

(6) $s_1 = 0, s_2 = 0 \Rightarrow x_1 = 2, x_2 = 4$

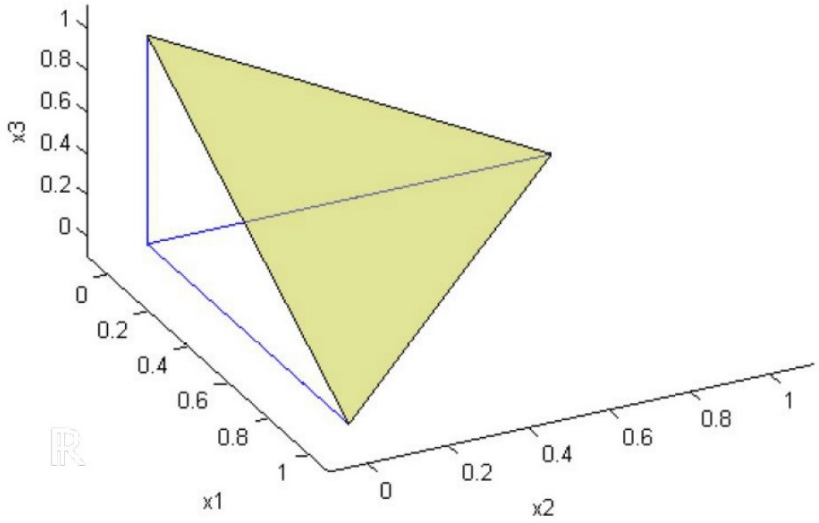
The intersections where some x_j or s_i are < 0 yield infeasible solutions.

Which are the vertices of a polyhedron in standard form?

Example

$$P = \{\underline{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1, \underline{x} \geq \underline{0}\}$$

with $n = 3$ and $m = 1$.



Property 3

For any polyhedron $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$:

1. *the facets (edges in $\mathbb{R}^2$) are obtained by setting one variable to 0,*
2. *the vertices are obtained by setting n-m variables to 0.*

In the above example: $3 - 1 = 2$ variables set to 0 for vertices.

Algebraic characterization of the vertices

Consider any $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$ in standard form.

Assumption: $A \in \mathbb{R}^{m \times n}$ is such that $m \leq n$ of rank m (A is of full rank).

Equivalent to assume that there are no "redundant" constraints.

Example

$$\begin{array}{rrrcl} 2x_1 & +x_2 & +x_3 & = & 2 \text{ (I)} \\ x_1 & +x_2 & & = & 1 \text{ (II)} \\ x_1 & & +x_3 & = & 1 \text{ (III)} \\ x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

Since (I) $\equiv$ (II) + (III), then (I) can be dropped.

- If $m = n$, there is a unique solution of $A\underline{x} = \underline{b}$ ($\underline{x} = A^{-1}\underline{b}$).
- If $m < n$, there are ∞ solutions of $A\underline{x} = \underline{b}$. The system has $n - m$ degrees of freedom, $n - m$ variables can be fixed arbitrarily. By fixing them to 0, we get a vertex.

$$P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$$

with n variables, m constraints, and $A \in \mathbb{R}^{m \times n}$.

Definition 14

A basis of such a matrix A is a subset of m columns of A that are linearly independent and form an $m \times m$ non singular matrix B.

$$A = [\overbrace{B}^{m} \mid \overbrace{N}^{n-m}]$$

First permute the columns of A, then partition A into $[B|N]$

Let $\underline{x} = [\ \overbrace{\underline{x}_B^T}^{m \text{ components}}\ |\ \overbrace{\underline{x}_N^T}^{n-m \text{ components}}\ ]^T$. Any system $A\underline{x} = \underline{b}$ can be written as $B\underline{x}_B + N\underline{x}_N = \underline{b}$. Since B is non singular, for any set of values for $\underline{x}_N$ we have

$$\underline{x}_B = B^{-1}b - B^{-1}N\underline{x}_N$$

Definition 15

- A basic solution is a solution obtained by setting $\underline{x}_N = \underline{0}$ and, consequently, letting $\underline{x}_B = B^{-1}\underline{b}$.
- A basic solution with $\underline{x}_B \geq 0$ is a basic feasible solution (BFS).
- The variables in $\underline{x}_B$ are the basic variables, those in $\underline{x}_N$ are the non basic variables.

Note: By construction $(\underline{x}_B^T, \underline{x}_N^T)$ together satisfy $A\underline{x} = \underline{b}$.

Theorem 16

$\underline{x} \in \mathbb{R}^n$ is a basic feasible solution if and only if $\underline{x}$ is a vertex of $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$.

Example

$$
\begin{array}{ll}
\min & 2x_1+x_2+5x_3 \\
\text{s. t.} & \begin{array}{rl}
x_1+x_2+x_3+x_4 & = 4 \\
x_1 \qquad\qquad +x_5 & = 2 \\
x_3 \qquad\quad +x_6 & = 3 \\
3x_2+x_3 \qquad\qquad +x_7 & = 6 \\
x_1,\ x_2,\ x_3,\ x_4,\ x_5,\ x_6,\ x_7 & \geq 0
\end{array}
\end{array}
\quad A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 3 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ 3 \\ 6 \end{bmatrix}
$$

- Choosing columns 4, 5, 6, 7, we have $B = I = B^{-1}$, so $\underline{x}_B = B^{-1}\underline{b} = \underline{b} \geq \underline{0}$, thus we obtain a basic feasible solution.
- Choosing columns 2, 5, 6, 7, we have

$$
B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{bmatrix}, x_B = \begin{bmatrix} 4 \\ 2 \\ 3 \\ -6 \end{bmatrix}
$$

and thus we obtain an infeasible basic solution.

Number of basic feasible solutions

At most one basic feasible solution for each choice of the $n - m$ non basic variables (to be set to zero) out of the n variables:

$$\#\text{ basic feasible solutions } \leq \binom{n}{n-m} = \frac{n!}{(n-m)!(n-(n-m))!} = \binom{n}{m}.$$